

CS391 – Entrepreneurship & Technology Commercialization

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Reference Book: *Entrepreneurship: Successfully Launching New Ventures*

6th Edition by Bruce R. Barringer & R. Duane Ireland

Week 1

Introduction to Entrepreneurship & Technology Commercialization

Lecture Overview

- Foundation concepts of **entrepreneurship, innovation, and technology commercialization**
- Why entrepreneurship matters for economic development
- How **technology ventures differ** from traditional businesses
- The **commercialization journey**

Learning Objectives (CLO_1)

By the end of Week 1, students should be able to:

- Define **entrepreneurship, innovation, and technology commercialization**
- Explain the role of entrepreneurs in **economic growth & technological advancement**
- Distinguish between **traditional vs. technology-based entrepreneurship**
- Describe the process of **turning a tech idea into a commercial product**
- Understand the **entrepreneurial ecosystem** (investors, incubators, govt., universities)

1. What is Entrepreneurship?

Definition:

The process of creating, managing, and growing a business venture to capture opportunities in the market.

Key Characteristics:

- Risk-taking
- Innovation
- Vision
- Adaptability
- Persistence

1. What is Entrepreneurship?

Types of Entrepreneurship:

- Small business entrepreneurship
- Scalable startup entrepreneurship
- Social entrepreneurship
- Corporate entrepreneurship (intrapreneurship)

2. Role of Entrepreneurship

- **Engine of innovation**
- **Job creation** and wealth generation
- Drives **competition** and improved quality of life
- **Examples:**
 - Silicon Valley startups (Google, Apple, Tesla) reshaping industries

3. Technology Commercialization

Definition:

Transforming research/innovation into marketable products or services.

Steps:

1. Idea / Research
2. Proof of concept / Prototype
3. Feasibility study
4. Business model development
5. Funding & scaling
6. Market launch & growth

3. Technology Commercialization

Why Different from Traditional Ventures?

- Higher risk, higher reward
- Needs **IP protection, licensing, spin-offs**
- Often depends on **VCs & angel investors**

4. Technology-Based Entrepreneurship

- Entrepreneurs leveraging **cutting-edge technologies** (AI, biotech, renewable energy, IoT).
- **Examples:**
 - SpaceX → Commercial space tech
 - Zoom → Communication platform
 - Bykea (Pakistan) → Mobile + logistics

Key Challenge:

Balancing **technical feasibility** with **market desirability**.

5. The Entrepreneurial Ecosystem

Components:

- Entrepreneurs, investors, mentors
- Universities, government, support networks

Role of Incubators/Accelerators:

- Y Combinator (US)
- Plan9 (Pakistan)

Government & IP Policies:

- Protect innovation
- Enable commercialization

Class Activities

Icebreaker Discussion (10 mins):

- “Who is an entrepreneur you admire and why?”

Think-Pair-Share (15 mins):

- Q: “What’s the difference between a small shop and a tech startup?”

Mini Workshop (20 mins):

- Groups receive 3 raw technologies (e.g., solar panel, AI chatbot, drone).
- Task: Convert into a business idea.
- Share ideas with class.

Case Studies

Global Example:

Apple iPhone → R&D to global commercialization

Local Example (Pakistan):

Careem → Ride-hailing, acquired by Uber

Failure Example:

Google Glass → Strong tech, failed commercialization due to **poor market fit**

Suggested Readings

From Reference Book (Barringer & Ireland):

- Chapter 1: *Introduction to Entrepreneurship*
- Chapter 3: *Feasibility Analysis*

Supplementary:

- Christensen, C. (1997). *The Innovator's Dilemma* (Intro chapter)
- University Tech Transfer articles (e.g., MIT, Stanford)

Mini Assignment (Due Week 2)

Task: Pick one technological innovation (local or global). Write a 1-page reflection:

1. What problem does it solve?
2. Why is it innovative?
3. How was it commercialized (sold/licensed/startup)?
4. What challenges did it face?

Purpose: Warm-up for Week 2 → Opportunity Recognition

Acknowledgement

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